

Project Planning

User Stories

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1. “As a busy professional with a passion for gardening, I want a greenhouse that adjusts environmental conditions—like temperature, humidity, and lighting—tailored to the specific needs of my plants. This way, even when my schedule doesn't allow for daily gardening attention, I can trust that my plants are thriving in an optimal environment, without my direct intervention.” - *Sophie, Nurse.*
2. “As someone who often forgets the watering schedule, I need a system that can reliably manage the irrigation of my plants. I want the greenhouse to monitor the soil moisture levels and water the plants as needed, ensuring they receive the right amount of water at the right time, safeguarding them from the consequences of my forgetfulness.” - *Oliver, Graphic Designer.*
3. “As an individual who is deeply committed to eco-friendly practices, I'm looking for a greenhouse solution that operates efficiently, using the least amount of electricity possible. I want a system that smartly conserves energy while maintaining the necessary conditions for plant growth, thereby minimizing my carbon footprint and aligning with my values of environmental stewardship.” - *Ava, Chef*
4. “As a novice gardener eager to delve into horticulture, I'm in search of a greenhouse system that simplifies the cultivation process. I need a setup that can maintain the right conditions automatically—managing factors like the temperature and the timing of the lights—so I can focus on the joys of gardening rather than the complexities. This system would give me hands-on experience while ensuring the plants get what they need, allowing me to learn and grow as a gardener at my own pace.” - *Ethan, Teacher.*

Preliminary Task List with Durations

User Story 1: Environmental Control - Sophie, Nurse

Preliminary Task List with Durations:

- Configure Raspberry Pi for Environmental Monitoring (3 days): Install and set up the operating system and prepare software for sensor integration.
- Install and Calibrate Sensors (2 days): Set up temperature, humidity, and light sensors within the greenhouse.
- Develop Environmental Control Logic (3 days): Program the Raspberry Pi to automatically adjust conditions based on sensor data.

Resources:

- Human: Web Development Students (for technical setup and programming), Plant Knowledge Coordinator (for advising on optimal environmental conditions based on plant care experience).
- Material: Raspberry Pi, Temperature/Humidity/Light Sensors, Relay Module

User Story 2: Automated Watering - Oliver, Graphic Designer

Preliminary Task List with Durations:

- Identify Optimal Soil Moisture Levels (2 days): Research and document moisture needs for various plant types.
- Install Soil Moisture Sensors and Water Pump (2 days): Physically set up the hardware in the greenhouse.
- Implement Watering Automation (3 days): Code the logic for automated watering based on sensor feedback.

Resources:

- Human: Web Development Students (for system programming and sensor integration), Plant Knowledge Coordinator (for input on watering schedules and moisture requirements).
- Material: Soil Moisture Sensors, Water Pump, Hose, Raspberry Pi

User Story 3: Energy Efficiency - Ava, Chef

Preliminary Task List with Durations:

- Evaluate and Select Energy-Efficient Components (2 days): Research and choose low-energy lights, fans, and other necessary hardware.
- Program Energy Conservation Features (3 days): Code the Raspberry Pi to reduce energy consumption, including implementing a sleep mode.
- Optimize System for Minimal Power Usage (2 days): Refine the software to ensure efficient operation throughout the greenhouse's daily cycle.

Resources:

- Human: Web Development Students (for implementing energy-saving solutions and component selection), Plant Knowledge Coordinator (ensuring energy-efficient practices align with plant growth needs).
- Material: Energy-efficient LED Lights, Energy-efficient Fan, Raspberry Pi

User Story 4: Simplified Cultivation - Ethan, Teacher

Preliminary Task List with Durations:

- Design User-Friendly System Interface (3 days): Create a straightforward interface for monitoring and controlling the greenhouse environment.
- Setup Environmental Automatic Controls (2 days): Program temperature and lighting schedules that cater to different plant growth stages.
- Compile Basic Gardening Tips (2 days): Gather and organize easy-to-understand gardening advice tailored to the system's capabilities.

Resources:

- Human: Web Development Students (for interface development and automation programming), Plant Knowledge Coordinator (for guidance on gardening basics and system usability from a gardener's perspective).
- Material: Raspberry Pi, Display/Interface Hardware

WBS for Eco-Friendly Smart Greenhouse Controller Project

Num	Task	Dur (Days)	Dep	Resource
1	Environmental Control for Sophie, Nurse	0		
1.1	Configure Raspberry Pi for Environmental Monitoring	3		Web Development Students
1.2	Install and Calibrate Sensors	2	1.1	Web Development Students, Plant Knowledge Coordinator
1.3	Develop Environmental Control Logic	3	1.2	Web Development Students
2	Automated Watering for Oliver, Graphic Designer	0		
2.1	Identify Optimal Soil Moisture Levels	2		Plant Knowledge Coordinator
2.2	Install Soil Moisture Sensors and Water Pump	2	2.1	Web Development Students
2.3	Implement Watering Automation	3	2.2	Web Development Students
3	Energy Efficiency for Ava, Chef	0		
3.1	Evaluate and Select Energy-Efficient Components	2		Web Development Students

Num	Task	Dur (Days)	Dep	Resource
3.2	Program Energy Conservation Features	3	3.1	Web Development Students
3.3	Optimize System for Minimal Power Usage	2	3.2	Web Development Students
4	Simplified Cultivation for Ethan, Teacher	0		
4.1	Design User-Friendly System Interface	3		Web Development Students
4.2	Setup Environmental Automatic Controls	2	4.1	Web Development Students
4.3	Compile Basic Gardening Tips	2	4.1	Plant Knowledge Coordinator